

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Task 3 Report Model Validation, Overfitting Control & Hyperparameter Tuning

1. Introduction

This task focuses on validating a machine learning regression model using the California Housing dataset. The objective is to detect overfitting, evaluate model performance using cross-validation, and improve accuracy through hyperparameter tuning.

2. Objectives

- Load and preprocess the dataset.
- Train a Decision Tree Regressor.
- Detect overfitting.
- Perform 5-fold cross-validation.
- Optimize using GridSearchCV.
- Compare with Linear and Ridge Regression.

3. Tools Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib and Scikit-learn.

4. Dataset

The California Housing dataset contains housing attributes such as median income, house age, average rooms, population, latitude and longitude. The target variable is the median house price.

5. Methodology

The dataset was standardized using StandardScaler and split into training and testing sets (80:20). A Decision Tree Regressor was trained and evaluated using RMSE. Cross-validation and GridSearchCV were then applied to reduce overfitting and determine the best hyperparameters. Finally, the optimized model was compared with Linear Regression and Ridge Regression.

6. Results

Metric	Value
Training RMSE	~0 (3.218e-16)

Testing RMSE	0.7030
Cross Validation RMSE	0.8957
Best max_depth	10
Best min_samples_split	10
Optimized RMSE	0.6454
R ² Score	0.6821

The optimized Decision Tree achieved the lowest RMSE (0.6454) and the highest R² score (0.6821), outperforming both Linear Regression and Ridge Regression (RMSE 0.7456, R² 0.5758).

7. Discussion

The initial Decision Tree model showed overfitting because the training error was almost zero while the testing error was significantly higher. GridSearchCV selected max_depth=10 and min_samples_split=10, reducing model complexity and improving generalization. Cross-validation provided a more reliable estimate of model performance.

8. Conclusion

The task successfully demonstrated model validation, overfitting detection, cross-validation, and hyperparameter tuning. The optimized Decision Tree model produced the best predictive performance and is the preferred model for this regression problem.

9. References

Scikit-learn Documentation
Python Documentation
California Housing Dataset